

Decide whether each of the following is statement. If it is a statement, decide whether or not it is compound.

1. Nashville is the capital of Tennessee.
2. Got milk?
3. China does not have a population of more than 1 billion, and it is a country.
4. If I get an "A" then I will celebrate.

Write a negation of each statement.

5. This is not the time to complain.
6. $y \leq 5$
7. p
8. $p \wedge q$

Fill in the following truth tables.

p	q	$p \wedge q$
T	T	
T	F	
F	T	
F	F	

p	q	$p \vee q$
T	T	
T	F	
F	T	
F	F	

Let p represent a false statement and q represent a true statement. Find the truth value of each compound statement.

9. $\sim p$
10. $p \vee \sim q$
11. $\sim (p \vee \sim q)$

Let p represent a true statement, and q and r represent false statements. Find the truth value of each compound statement.

12. $q \vee \sim r$

13. $(\sim p \wedge q) \vee \sim r$

Let p be the statement " $2 > y$ ", q be statement " $8 \leq 6$," and r be the statement " $19 \leq 19$." Find the truth value of each compound statement.

14. $\sim p \vee \sim r$

15. $(p \wedge q) \vee \sim r$

16. $(\sim r \wedge q) \vee \sim p$